VALHEIM COASTAL MEGA-BASE — MASTER PLAN & CONSTRUCTION MANUAL

Authoritative Architectural Dossier • 4.0 m Grausten Grid • Origin (0, 0, +4.0 m) at SW Corner of 24 × 24 m Dreamhouse

1. Requirements Analysis, Reference Audit & Resolved Contradictions

This master plan translates the project specification (`VALHEIM-COASTAL-MEGA-BASE.md`) and the two reference builds by Crate Of Bananas—Version 1: My Valheim Dream House (Stone & Wood) and Version 2: Dream House with Ashlands Materials—into a dimensionally rigorous, buildable coastal settlement in the Meadows biome. Every fixed constraint is preserved without exception: the existing **24 m × 24 m Dreamhouse** remains at `X = 0..+24 m, Y = 0..+24 m` with its main floor at `Z = +4.0 m` and roof finial at `Z = +28.0 m` (24 m building height); the first outdoor **36 m × 36 m Elevated Plaza** sits at `Z = +2.0 m`; and the primary settlement ground datum sits at `Z = 0.0 m`.

Before laying out the site, six technical tensions between the reference videos, vanilla Valheim building mechanics, and the written specification were identified and resolved:

Technical Issue / Tension	Reference Video / Game Mechanic	Authoritative Engineering Resolution
1. 36×36 m Plaza vs. 24×24 m House on a 4 m Grid	24 m equals 6 tiles of Grausten Floor 4×4 m; 36 m equals 9 tiles. Equal centering leaves 6 m (1.5 tiles) per side, breaking the 4 m grid by 2 m.	Offset the 36×36 m Plaza to `X = -4..+32 m, Y = -8..+28 m`. Every corner falls on an exact multiple of 4.0 m, creating an 8 m seaward forecourt (South), 8 m comfort terrace (East), 4 m bridge cloister (West), and 4 m service gallery (North).
2. Basement Height (3 m Video vs. 4 m Spec)	In Version 2 (08:20), the creator builds a 3 m basement on a 40×40 m pad. Section 2.1–2.2 mandates a +4.0 m main floor and +2.0 m plaza (36×36 m).	Enforce `Z = +4.0 m` main floor and `Z = +2.0 m` plaza per Section 2.1. Two courses of Grausten Wall 4×2 (2.0 m high each) yield an exact 4.0 m basement from `Z = 0.0 m` to `+4.0 m` without awkward beam shims.
3. Cardinal Exits vs. Seaward Primary Facade	In Version 1 (00:19, 12:18), 3 sides have doors and 1 side is blocked by the dual-hearth kitchen chimney.	Orient South (`-Y`) toward the sea and place the dual-hearth chimney on the North (inland) wall (`X = 8..16 m, Y = 22..25.5 m`). This leaves the South (Seaward), West (Resting Hall), and East (Plaza/Farm) portals 100% unobstructed.
4. Darkwood Piece Inventory vs. Wall Panels	Verified Darkwood Structures and Shingle Roofs include beams, poles, arches, carved dividers, gates, and roofs—but no solid wall panels.	Use Black Marble and Grausten masonry for lower plinths and structural walls, framed with Darkwood 2 m/4 m beams, arches, and Ashwood Walls / Carved Darkwood Dividers for upper timber infill.

Technical Issue / Tension	Reference Video / Game Mechanic	Authoritative Engineering Resolution
5. Continuous 4 m Paving vs. Crop Mechanics	Per Cultivator mechanics, crops require raw cultivated soil, 1.0 m spacing (3.6 m for Vineberries), and 100% unobstructed sky.	Design four recessed 12 m × 12 m open-sky soil beds (`Z = -0.15 m`) curbed by Grausten beams inside 4 m paved walkways, with an unroofed East masonry wall for climbing Ashvines.
6. Harbor Ship Sizing & Lox/Asksvin Containment	A Longship is 22×6 m; a Drakkar is ~32×10 m. Tamed Asksvin and Lox break stone walls when aggravated and only breed within 64 m of the player.	Provide a 36 m × 20 m harbor basin dredged to `Z = -4.5 m` (3.0 m water depth below `-1.5 m` sea level) and house Lox/Asksvin in a `Z = -2.0 m` sunken earth-core paddock 32–58 m NE of the Dreamhouse.

2. Comparative Evaluation of Two Master-Plan Concepts

Two distinct architectural layouts were developed that both preserve the fixed Dreamhouse and three-level elevation sequence. **Concept A (The Seaward Axial Forum & Sheltered Quay)** organizes the settlement along a triumphal North–South spine (`X = +8..+16 m`) with a continuous 8 m cart circulation loop, attaching the Residential/Resting Extension to the West via a level `Z = +4.0 m` skybridge while grouping Agriculture and Animal Husbandry on the East/Northeast downwind flank. **Concept B (The Terraced Acropolis & Twin-Pier Seafront)** attempts strict bilateral symmetry with twin diagonal harbor piers and flanking agricultural/animal wings directly abutting the central plaza.

Evaluation Criterion	Concept A: Seaward Axial Forum & Sheltered Quay (Selected)	Concept B: Terraced Acropolis & Twin-Pier Seafront
4 m Grid & Plaza Geometry	**100% Integer 4 m Alignment:** Asymmetric `-4..+32 m` plaza keeps every building and road on exact 4×4 m Grausten tiles.	**2 m Half-Tile Misalignment:** Centering a 24 m house on a 36 m plaza forces 6 m (1.5-tile) side margins, breaking the 4 m grid.
Visual Impact & Hierarchy	**Unbroken Seaward Vista:** 8 m axial avenue frames the Dreamhouse above the harbor; subordinate 16 m Resting Hall preserves anchor dominance.	**Split Focal Point:** Twin piers divide the waterfront view and flanking barns compete visually with the central plaza.
Circulation & Cart Logistics	**4.0 m Indoor Bridge + 8 m Loop:** 0.6-sec level indoor walk to beds; straight 46 m cart haul from quay to storage core.	**Detour Around Plinths:** Requires descending to 0 m and climbing back to +4 m to reach sleeping quarters or hauling carts up switchbacks.
Harbor Integration	**Single 36×20 m Deep Basin:** Parallel mooring displays the Longship/Drakkar on the main axis and allows easy turn-around.	**Cramped Twin Finger Piers:** Narrow slips hinder Drakkar turning and double the breakwater stone cost.

Evaluation Criterion	Concept A: Seaward Axial Forum & Sheltered Quay (Selected)	Concept B: Terraced Acropolis & Twin-Pier Seafront
Husbandry Safety & Buffs	**Protected Yet Active (28–58 m):** NE sunken pit keeps Lox/Asksvin within the 64 m breeding radius but far from fragile house walls.	**High Raid Collateral Risk:** Flanking pens sit 8 m from the main plaza, risking Lox/Asksvin smashing the house during raids.
Buildability & Expansion	**Modular 9-Stage Phasing:** Clean rectangular terraces reduce hoe/World Edit leveling time; easy expansion East or West.	**Complex Diagonal Retaining Walls:** Angled piers require heavy overlapping of rectangular Grausten pieces.

Recommendation & Architectural Justification: Concept A is selected because it resolves the fundamental parity mismatch between the 6-tile (24 m) Dreamhouse and the 9-tile (36 m) Elevated Plaza without breaking the 4.0 m Grausten grid. By giving the Plaza an 8.0 m (2-tile) seaward forecourt on the South (`Y = -8..0 m`) and an 8.0 m terrace on the East (`X = +24..+32 m`) while keeping a 4.0 m (1-tile) gallery on the West (`X = -4..0 m`) and North (`Y = +24..+28 m`), the West Skybridge spans only 4.0 m to connect the Dreamhouse main floor (`Z = +4.0 m`) directly to the Resting Hall (`X = -20..-4 m, Z = +4.0 m`).

3. Authoritative 2D Master Site Plan & Coordinate System

The selected **Concept A Master Plan** occupies a leveled land envelope of **128 m (East–West, `X = -44..+84 m`) × 116 m (South–North, `Y = -44..+72 m`)** plus a **28 m seaward harbor zone (`Y = -72..-44 m`)**, totaling **128 m × 148 m (32 × 37 Grausten 4×4 m tiles)**. Origin `(0, 0, 0)` is locked at the southwest corner of the Dreamhouse footprint at ground datum.



Figure 1: Deliverable 1 — Dimensionally accurate 2D Master Site Plan (Drawing VMB-MP-01) showing the 4.0 m Grausten grid, coordinate axes, zone boundaries, building footprints, and fixed +4.0 m / +2.0 m / 0.0 m elevation hierarchy.

Complete Zone & Building Coordinate Schedule

Code & Building Name	Functional Role	Footprint (m)	X Range (m)	Y Range (m)	Elev Z (m)	Primary Materials & Notes
A1. Existing Dreamhouse	4 Crafting Stations, 60 Chests, Kitchen, Basement Upgrades	24 × 24 m (H: 24 m)	`0 to +24`	`0 to +24`	Floor `+4.0` Base `0..+4`	Preserved anchor; 8 grounded Flametal/Iron columns at `Z=0.0m`; North chimney at `Y=22..25.5m`.
A2. First Elevated Plaza	Primary ceremonial terrace & 3-level circulation hub	36 × 36 m	`-4 to +32`	`-8 to +28`	`+2.0 m`	Fixed 9×9 Grausten 4×4m grid; Black Marble retaining walls; 4 cardinal staircases.
A3. Resting Hall & Bridge	Max-Comfort (18+) sleeping quarters & mead hall	16 × 16 m + 4×8 m	`-20 to -4` Bridge `-4..0`	`+8 to +24` Bridge `+8..+16`	`+4.0 m` (Peak `+16m`)	Level indoor skybridge from Dreamhouse West door; Darkwood frame, Shingle roof, Dragon/Ashwood beds.
A4. Spa & Hot Tub Court	Outdoor Comfort terrace (Hot Tub, Hearth, Maypole)	20 × 8 m	`-24 to -4`	`0 to +8`	`+2.0 m`	Directly adjacent to Dreamhouse SW corner & Resting Hall so Comfort buffs overlap beds.
B1–B4. Crop & Orchard Beds	Carrots (B1), Turnips (B2), Onions (B3), Orchard/Bees (B4)	4× (12 × 12 m)	`+44 to +72`	`-28 to 0`	`-0.15 m` (Walks `0.0m`)	Unroofed raw cultivated soil (~360 crops at 1.0m spacing + 12 Beehives & 4 trees at 4–6m spacing).
B5–B6. Granary & Ashvines	Seed chests, Cultivator bench, Windmill & Vineberry wall	28 × 8 m + 2×32 m	`+44 to +72` Wall `+74..+76`	`+4 to +12` Wall `-28..+4`	`0.0 m` (Tower `+13m`)	East Windmill catches coastal breeze; 8 Ashvine clusters spaced at 4.0m (>3.6m min) on East wall.

Code & Building Name	Functional Role	Footprint (m)	X Range (m)	Y Range (m)	Elev Z (m)	Primary Materials & Notes
C1. Boar & Hen Barn	Roofed breeding pens (protected from Bat/Drake raids)	12 × 16 m	`+44 to +56`	`+28 to +44`	`0.0 m`	Double-gated airlock entry; Grausten lower walls, Darkwood trusses, enclosed poultry loft.
C2. Lox / Asksvin Paddock	Heavy mount containment & breeding pit	12 × 24 m	`+60 to +72`	`+28 to +52`	`-2.0 m` (Wall `+2.0m`)	Indestructible 2m sunken earth core faced with Black Marble; 4m riding ramp at North end.
C3. Saddle & Feed House	Animal feed storage, saddles, butcher knife & egg warmers	12 × 6 m	`+44 to +56`	`+46 to +52`	`0.0 m`	Directly services both C1 Barn and C2 Paddock; sits 42m from Dreamhouse (inside 64m active zone).
D1–D3. Defense & Perimeter	North Barbican Gatehouse, Curtain Wall, 4 Bastions, Moat	128 × 116 m (Gate 16×10m)	`-44 to +84` Gate `+4..+20`	`-44 to +72` Gate `+64..+74`	Wall `+3.0 m` Moat `-2.5 m`	4m wide × 2.5m deep ha-ha ditch stops ground raid pathing without blocking ocean views.
E1–E3. Small Harbor & Quay	36m Quay, Stepped Gangway, Warehouse, Breakwater Beacon	36 × 20 m Basin 36 × 4 m Quay	`-24 to +28`	`-72 to -36`	Quay `0.0/-0.5m` Bed `-4.5 m`	Waterline at `Z=-1.5m` gives 1.5m quay freeboard and 3.0m clear water draft for Longship/Drakkar.
F1–F3. Portals & Public Spaces	6-Bay Stone Portal Colonnade, Sacred Oak, Obliterator, Obelisk	20 × 10 m + 10 × 10 m	`-28 to -8` SW `-32..-22`	`+38 to +48` SW `-30..-20`	`0.0 m`	Grausten Medium Arches frame Ashlands Stone Portals; SW Obliterator isolated from timber roofs.

4. Vertical Sections, Elevation Transitions & Circulation Engineering

To ensure zero ambiguity during in-game construction, Figure 2 documents Longitudinal Section A–A (South–North through `X = +12.0 m`), Transverse Section B–B (West–East through `Y = +12.0 m`), the complete stair/ramp schedule, and the 20 m radius spawn-suppression grid. Because standard Grausten Stairs and Stone Stairs each rise `1.0 m` over a `2.0 m` horizontal run, every `2.0 m` elevation step

(`0.0 m → +2.0 m` and `+2.0 m → +4.0 m`) uses exactly two stair flights placed in series (`4.0 m` total run = 1 Grausten 4×4 m grid tile), keeping all stair landings flush with the 4.0 m module.

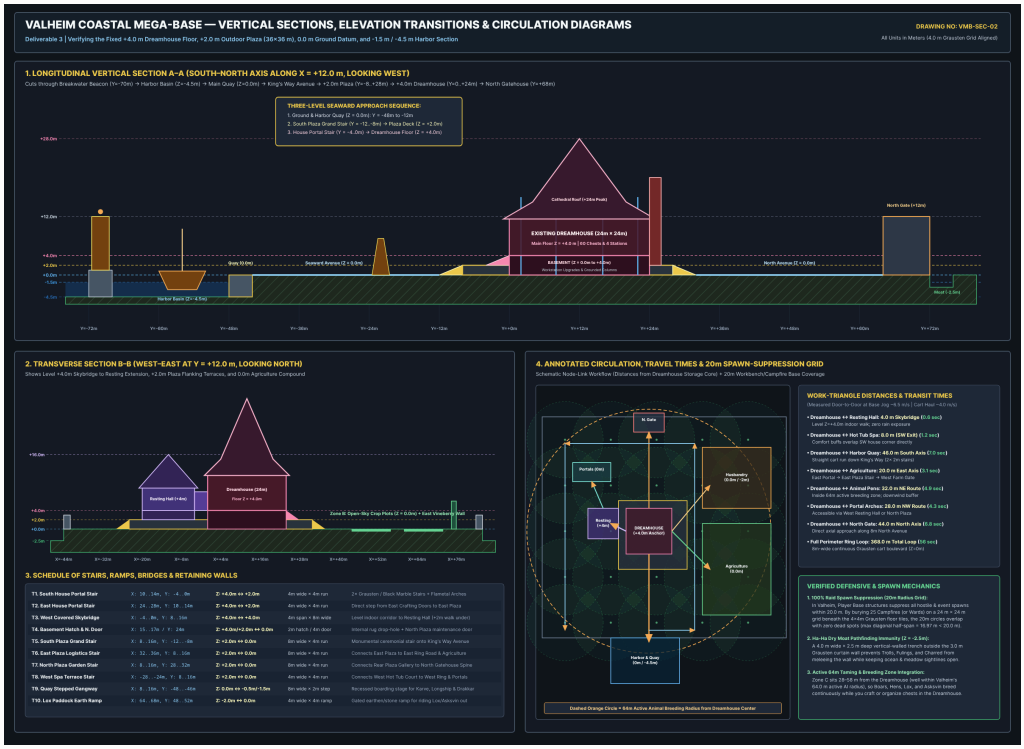


Figure 2: Deliverable 3 — Vertical Sections A–A and B–B, Stair/Bridge Schedule, Work-Triangle Transit Times, and 20 m Campfire/Workbench Spawn-Suppression Grid (Drawing VMB-SEC-02).

Structural Integrity & Grounded Column Verification: Per verified Valheim structural stability mechanics, stone and Grausten pieces lose 1/10 support vertically and 1/3 horizontally, whereas Wood-Iron and Flametal poles rise up to 50 m vertically when anchored directly to terrain (Blue stability). Therefore, during Stage 1 earthworks, the `24 m × 24 m` Dreamhouse basement floor must remain at `Z = 0.0 m` natural/leveled terrain so that the 8 primary Flametal / Wood-Iron columns extend from `Z = 0.0 m` through the `+4.0 m` main floor to the `+16.0 m` roof ring-beam with 100% grounded support.

5. Phased 9-Stage Construction Schedule & In-Game Execution Playbook

Follow this exact 9-stage sequence to avoid structural collapses, prevent rain decay on exposed wood, and eliminate terrain-leveling rework around finished masonry:

Stage & Sequence	Scope & Coordinate Target	Practical In-Game / World Edit Execution Steps
Stage 1: Site Leveling, Dredging & Spawn Suppression	Full site `X: -48..+88 m, Y: -76..+76 m`	1. Establish Origin `(0,0,0)` at the SW corner of the Dreamhouse foundation. 2. Flatten the main settlement plateau (`X: -44..+84 m, Y: -44..+72 m`) to `Z = 0.0 m`. 3. Raise the 36×36 m Plaza earth core (`X: -4..+32 m, Y: -8..+28 m`) and West Spa core (`X: -24..-4 m, Y: 0..+24 m`) to `Z = +2.0 m`, leaving the 24×24 m Dreamhouse basement cavity at `Z = 0.0 m` so its 8 internal columns remain grounded in earth (Blue stability). 4. Excavate the 4 m wide Perimeter Ha-Ha Moat to `Z = -2.5 m`, the NE Lox Paddock (`X: 60..72 m, Y: 28..52 m`) to `Z = -2.0 m`, and dredge the Harbor Basin (`X: -8..+28 m, Y: -68..-48 m`) to `Z = -4.5 m`. 5. Bury 25 Campfires on the 24 m × 24 m grid (`X = -32, -8, +16, +40, +64 m`; `Y = -36, -12, +12, +36, +60 m`) and level soil over them with the Hoe to lock in 100% hostile spawn suppression.
Stage 2: Master 4 m Grausten Grid & Datum Curbs	All paved zones on 4.0 m module	Snap the first master Grausten Floor 4×4 tile at `(0,0)` aligned to true North–South / East–West compass axes. Extend 4×4 m reference ribbons along `X = 8..16 m` (King's Way) and the `+2.0 m` Plaza perimeter (`X: -4..+32 m, Y: -8..+28 m`). Leave the four 12×12 m agricultural plots (`B1–B4`) unpaved so soil remains exposed to the sky.
Stage 3: Dreamhouse Surroundings & +2.0 m Plaza	`X: -4..+32 m, Y: -8..+28 m, Z: +2.0 m`	Face the 36×36 m Plaza perimeter with 2.0 m high Black Marble plinth walls and Grausten Tapered Pillars at 4 m intervals. Construct the South House Portal Stair (`X: 10..14 m, Y: -4..0 m`) and East House Portal Stair (`X: 24..28 m, Y: 10..14 m`) connecting `Z = +4.0 m` to `+2.0 m`, then build the four 8 m wide Plaza Grand Stairs descending from `Z = +2.0 m` to `0.0 m`.
Stage 4: Primary Seaward Avenue & 8 m Ring Loop	Axis `X: 8..16 m`; Ring Loop `368 m`	Pave the 8 m wide Seaward Avenue (`Y = -44..-12 m`), Central Forecourt (`X = 0..24 m, Y = -28..-16 m` with the 7.5 m Black Marble Wayfinder Obelisk at `X = 10..14 m, Y = -24..-20 m`), North Avenue (`Y = 32..68 m`), and the 8 m wide continuous cart Ring Loop with Grausten 4×4 m tiles edged by Grausten Medium Beams.
Stage 5: Connected Resting Extension, Skybridge & Spa	`X: -24..-4 m, Y: 0..+24 m, Z: +2.0 / +4.0 m`	Erect the 4 m span × 8 m wide Covered Skybridge (`X: -4..0 m, Y: 8..16 m, Z: +4.0 m`) directly from the Dreamhouse West Portal using Wood-Iron / Flametal core beams clad in Darkwood. Build the 16×16 m Resting Hall (`X: -20..-4 m, Y: 8..24 m, Z: +4.0 m`) roofed first with 26°/45° Shingle Roofs to prevent rain damage, then furnish with Dragon/Ashwood Beds, Hearth, Throne, Rugs, and Banners. Complete the `Z = +2.0 m` Spa Court (`Y = 0..8 m`) with Hot Tub and Maypole within 10 m of the beds.

Stage & Sequence	Scope & Coordinate Target	Practical In-Game / World Edit Execution Steps
Stage 6: Small Harbor, 36 m Quay & Breakwater	`X: -24..+28 m, Y: -72..-36 m`	Build the continuous Black Marble seawall along `Y = -44 m` (top at `Z = 0.0 m`, 1.5 m above the `-1.5 m` waterline). Extend the 36 m × 4 m Main Quay (`X: -8..+28 m, Y: -48..-44 m`) with a recessed `Z = -0.5 m` stepped boarding stage at `X = 8..16 m`. Erect the 16×8 m Harbor Warehouse (`X: -24..-8 m, Y: -44..-36 m`) and the outer Breakwater Mole (`Y = -72..-68 m`) topped by the 12 m Beacon Tower at `X = 16..20 m`.
Stage 7: Agriculture & Animal Husbandry Compounds	East Wing `X: +40..+76 m, Y: -32..+56 m`	Enclose Zone B (`Y = -32..+16 m`) with 2 m Grausten window walls. Frame the four 12×12 m soil plots (`B1`–`B4`) with Grausten Small Beams, cultivate the soil with the Cultivator, and build the 28×8 m Granary & Windmill (`B5`) plus the 6 m high East Vineberry Wall (`B6`). In Zone C (`Y = +24..+56 m`), build the roofed 12×16 m Boar/Hen Barn (`C1`), face the `-2.0 m` sunken Lox/Asksvin pit (`C2`) with Black Marble over its indestructible earth core, and add the 12×6 m Feed/Saddle Shed (`C3`).
Stage 8: Defensive Perimeter, Bastions & Gatehouse	Perimeter `X: -44..+84 m, Y: -44..+72 m`	Construct the 3.0 m high Grausten & Black Marble crenellated curtain wall along the inner edge of the `-2.5 m` dry moat, anchored by the four 8×8 m corner watchtowers (`Z = +10.0 m`) and the 16×10 m Twin-Tower Northern Barbican Gatehouse (`X: 4..20 m, Y: 64..74 m, Z = +12.0 m`) fitted with double Flametal/Darkwood gates.
Stage 9: Portal Sanctuary, Sacred Grove & Lighting	NW Grove `X: -36..-4 m, Y: +28..+52 m`	Erect the 20×10 m Grausten Arch Colonnade (`X: -28..-8 m, Y: 38..48 m`) housing 6 Stone Portals, plant the central Ancient Oak in the reflection court (`X = -18 m, Y = +32.5 m`), install the Lightning Obliterator on the SW bastion (`X = -27 m, Y = -25 m`), and place Standing Blue Iron Torches and Lava Lanterns at 8 m intervals along all avenues.